layer formation which depends to a great extent on the pH. From this result new experimental conditions may be established in order to solve practical problems concerning selective etching.

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Catalytic Activity of Iron, Nickel, and Nickel-Phosphorus in Electroless Nickel Plating

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ABSTRACT

The catalytic activity for electrooxidation of hypophosphite and electroless nickel plating on iron, nickel, and electrolessly plated nickel (with 2.2-2.9 weight percent (w/o) P) was investigated in an ammoniacal solution of pH 8.8 at 50°C by potential measurements and linear sweep voltammetry from −0.3V to 0.92V *vs.* SCE. Cathodic polarization of any of the substrates (in 0.1M H_2SO_4) before testing or permeation of hydrogen through iron foils during testing reduced the incubation time for electroless plating, increased anodic dissolution of the substrates in the passive region, and increased the hypophosphite electrooxidation on nickel and nickel-phosphorus at potentials of thermodynamic stability for nickel oxides or hydroxides. The nickel-phosphorus substrate exhibited significantly higher activity than pure nickel, in a similar manner to the effect of cathodic polarization. The effect of cathodic polarization and hydrogen is associated with the electrochemical reduction of surface oxides, whereas the higher activity of nickel-phosphorus deposits (as compared to that of pure nickel) is explained by the lower protectivity of surface oxides on these deposits.

Electroless nickel plating from hypophosphite solutions proceeds spontaneously on many metal surfaces (1-5), although the specifics of this process may be of a different nature. Some noble metals (palladium, rhodium, nickel, cobalt, gold) possess an intrinsic catalytic ability to induce electrooxidation of hypophosphite, and electroless nickel plating; whereas some active metals become catalytic due to the deposition of nickel from a solution by a displacement reaction. The latter phenomenon has been shown for iron, aluminum. and beryllium (2), gallium and thallium (6), magnesium, zinc, and manganese. The

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Key words: electroless, nickel plating, iron, nickel, nickel-phosphorus.

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catalytic activity of copper has been shown to be strongly affected by surface oxides (7). The activation of copper or its alloys by anodic treatment (4) or by the addition of chloride ions to a given plating solution (8) is evidently associated with the removal of these surface oxides.

Nickel and iron induce spontaneous electroless plating of nickel, but the presence of surface oxides may considerably prolong incubation times and decrease the adhesion of the deposits to the substrates. Investigations by Gaigher and van Wyke (9) showed that contamination of a nickel substrate due to exposure to laboratory air resulted in a blocky discontinuous growth of nickel electrodeposits, in contrast to smooth monolayer-type growth for uncontaminated surfaces.

The catalytic activity of a nickel surface for dehydrogenation of hypophosphite diminishes with time (2, 3) in the absence of Ni^{2+} ions. However, it is sustained in the presence of these ions, even at concentrations as low as 2×10^{-3}M (2). Gutzeit (2) indicated that the concentration of Ni^{2+} ions in the solution remains constant during a given experiment. Nevertheless, it seems that a small amount of deposition of electroless nickel is also possible, giving rise to the continuation of the catalytic activity.

Epifanova *et al.* (10) showed that the catalytic activity of electroless nickel for oxidation of hypophosphite was distinctly higher than that of pure nickel. This was explained in terms of the catalytic action of phosphorus. It is possible that the higher catalytic activity of nickel-phosphorus deposits when compared to that of nickel may also be due to other reasons, such as less vulnerability to oxide formation or other surface contamination, or to hydrogen absorbed by the deposits during plating. The effect of surface conditions on growth of the deposits was stressed by Cavallotti and Salvago (6). Inhibition of electroless plating and of hypophosphite electrooxidation by oxy-anions, oxides, or dissolved oxygen has been reported by several authors (7, 11-13).

Hydrogen has been shown to accelerate electroless plating, both by participation in the reduction of Ni^{2+} ions (1-3, 14), and also by reduction of surface oxides (13). On the other hand, hydrogen may diminish the catalytic activity of substrates, as it was observed for palladium (13).

The present work was undertaken in order to determine the effect of surface oxides and of hydrogen on the incubation time for electroless plating and on the catalytic activity for electrooxidation of hypophosphite on substrates of iron, nickel, and nickel-phosphorus.

Investigations were carried out in an ammoniacal hypophosphite solution, in which electroless plating produced nickel-phosphorus deposits with 2.2-2.9 w/o P (13). The catalytic activity and electrochemical behavior of the substrates were examined by potential measurements and linear sweep voltammetry.

Experimental

Tests were carried out on annealed Armco iron, 99.9% pure nickel, and preformed electroless nickel-phosphorus deposits containing 2.2-2.9 w/o P.

Specimens of Armco iron were electropolished in a solution of 940 ml acetic acid (glacial) + 60 ml perchloric acid, at 40-60V, while specimens of nickel were electropolished in a solution of 100 ml H_2SO_4 + 75 ml H_2O, at 5-6V; both at room temperature. Specimens were subsequently washed in distilled water, dried in air, and then masked with (Microshield™) lacquer to obtain an uninsulated surface area of 1 cm^2. The lacquer remained adherent and stable for periods of approximately 2 hr. Nickel-phosphorus deposits of ~3 μm thickness were grown on electropolished nickel by immersion into a plating solution (cf. Table I) at 50°C for 30 min.

Table I. Composition of electrolytes

Solution designation	$NaH_2PO_2 \cdot H_2O$ (g/liter)	$NiSO_4 \cdot 6H_2O$ (g/liter)	KNa tartrate $KNaC_4H_4O_6 \cdot 4H_2O$ (g/liter)	28% NH_4OH (ml/liter)	Conc H_2SO_4 (ml/liter)
Plating solution	13.25 (0.125 mol/liter)	13.1 (0.05 mol/liter)	28.1	70	*ca.* 15
Supporting solution	—	—	28.2	70	*ca.* 15
Supporting solution with Ni^{2+} ions	—	13.1	28.2	70	*ca.* 15
Supporting solution with sodium hypophosphite	13.25	—	28.2	70	*ca.* 15

Specimens were examined under the following conditions: (*i*) electropolished (iron or nickel) or freshly deposited (nickel-phosphorus), tested within 1 hr after preparation. (*ii*) As (*i*), followed by cathodic polarization in 0.1M H_2SO_4 at a current density of 100 $A \cdot m^{-2}$ for 10 min and tested immediately after polarization and rinsing with distilled water. (*iii*) As (*i*), aged in air at 50°C for 28 hr. (*iv*) As (*i*), immersed into 0.1M NaOH at 25°C, for 30 min. (*v*) Electropolished iron with continuous permeation of hydrogen.

A schematic drawing of a specimen instrumented for hydrogen permeation is shown in Fig. 1. A 0.10 mm thick membrane of annealed Armco iron was glued to a glass tube which then was filled with 0.1M H_2SO_4. The outer portion of the membrane served as the specimen's reaction surface. Permeation of hydrogen through this surface was achieved by cathodic polarization of the inner surface at a current density of 100 $A \cdot m^{-2}$. The rate of hydrogen permeation through Armco iron membranes was determined electrochemically (15) in separate experiments. In these experiments, the diffusion sides of the membranes were electroplated with palladium and the membranes were clamped between the two compartments of the electrolytic permeability cell. The diffusion compartment contained 0.02M NaOH with a reference electrode of Hg/HgO/0.02M NaOH. A typical current transient for hydrogen permeation during cathodic polarization of a membrane is shown in Fig. 2. No hydrogen gas bubbles were observed during the permeation experiments.

Specimens were examined in the solutions listed in Table I, prepared from reagent grade chemicals and distilled water. Sulfuric acid or, if necessary, NH_4OH, was added in amounts appropriate to adjust the pH of the solutions to 8.8. Measurements were carried out on stationary specimens at 50.0 ± 0.5°C in solutions deaerated by purging with pure nitrogen.

Measurements involved monitoring of open-circuit electrode potentials, and of linear sweep voltammograms. Potentials were measured against saturated calomel electrode (SCE) at 25°C via a Luggin capillary. Voltammetric measurements were performed using single cathodic sweeps at a rate of 1 mV/sec, beginning at −0.3V *vs.* SCE.

Results

Open-circuit potentials.—Changes of electrode potential for the substrates after immersion into electroless plating solutions are shown in Fig. 3 and 4. The rest potential of the electroless plating was approximately −0.87V (7, 13). The time required to reach this potential is defined as the incubation time and characterizes the ability of the substrates to become active for electroless plating.

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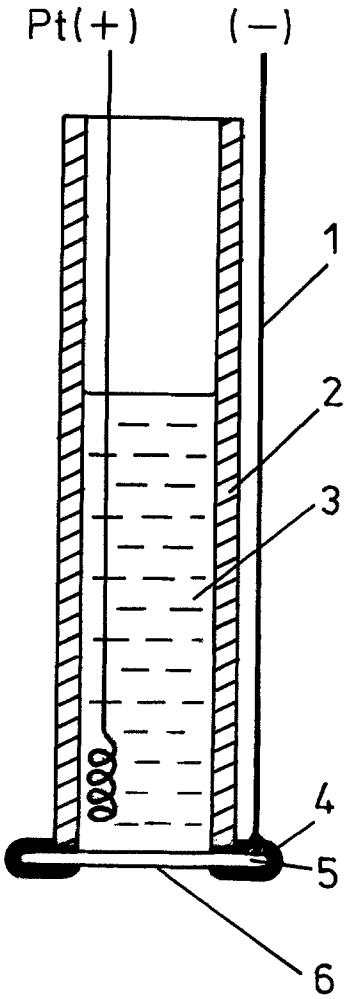


Fig. 1. Schematic drawing of a specimen unit for continuous permeation of hydrogen. (1) insulated wire, (2) glass tube, (3) 0.1M H_2SO_4, (4) epoxy resin, (5) 0.10 mm thick iron membrane, (6) specimen surface.

Figure 3 shows that electropolished iron becomes active after achieving a potential arrest at approximately −0.5V. These potential arrests virtually disappeared when specimens were cathodically polarized in 0.1M H_2SO_4 before testing. Permeation of hydrogen through iron membranes also diminished the extent of the potential arrest.

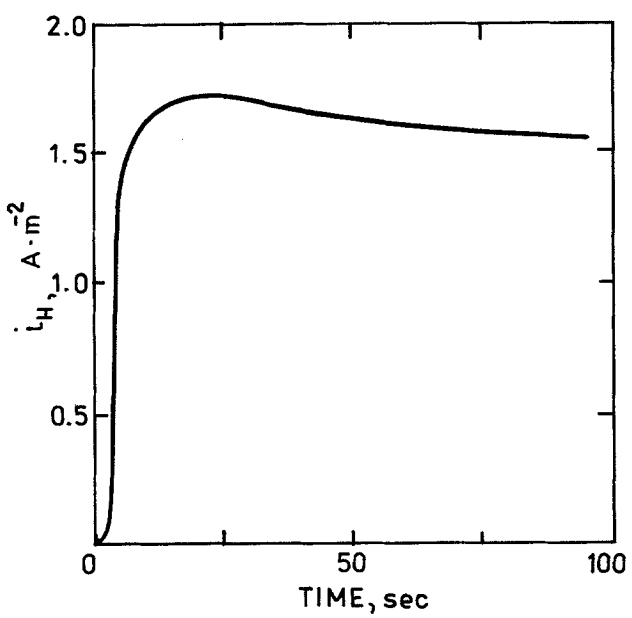


Fig. 2. Permeation of hydrogen (i_H) through a 0.10 mm thick iron membrane during cathodic polarization at 100 A · m^{-2} in 0.1M H_2SO_4, 50°C.

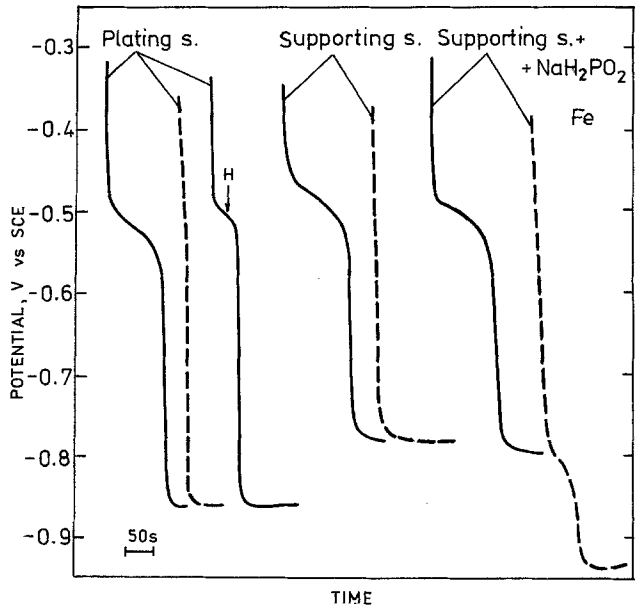


Fig. 3. Open-circuit potential of electropolished iron (full lines) and of iron after cathodic polarization in 0.1M H_2SO_4 (dashed lines) upon immersion into the plating, supporting, and supporting + NaH_2PO_2 solutions. An arrow with "H" indicates the start of hydrogen permeation through an iron membrane.

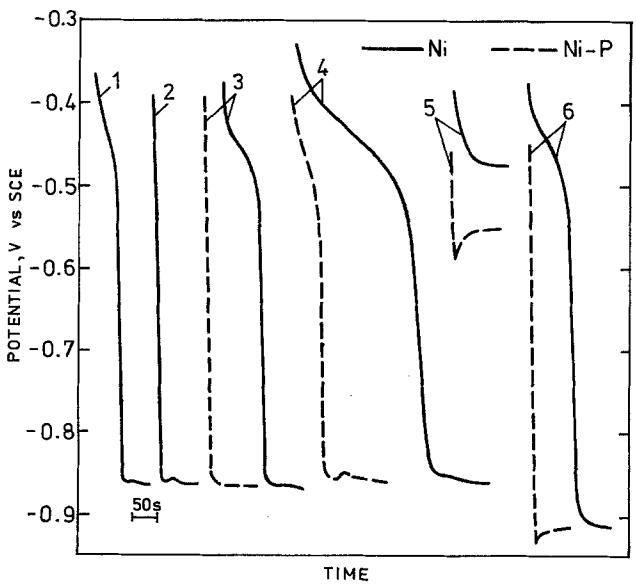


Fig. 4. Open-circuit potential of nickel and nickel-phosphorus deposits after immersion into the plating solution (curves 1-4), supporting solution (5) and supporting with NaH_2PO_2 (6): (1) electropolished nickel, (2) nickel after cathodic polarization, (3) after exposure to air at 50°C for 28 hr, (4) after exposure to 0.1M NaOH for 0.5 hr.

In the supporting solution (Table I), the rest potential of iron was approximately −0.78V. With a hypophosphite addition, the potential was nearly the same on an electropolished surface, but on cathodically polarized surfaces it shifted to approximately −0.92V, after a brief arrest at −0.78V.

The incubation time for plating on electropolished nickel (Fig. 4) was shorter than that for iron, and was also substantially diminished by cathodic prepolarization. Potential arrests were observed at approximately −0.45V.

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On nickel-phosphorus deposits, the potential arrests observed at −0.45V were significantly shorter than those observed on nickel, indicating higher activity of these deposits for the initiation of electroless plating. In solutions without Ni^{2+} ions, potentials on nickel-phosphorus dropped to lower values than on nickel.

Voltammetric measurements.—Measurements were made in the supporting solution with and without the additives essential for plating (cf. Table I), by scanning the potential in the cathodic direction from −0.3 to −0.92V or lower. Potential scanning was initiated immediately upon immersion.

Voltammograms on electropolished and cathodically polarized iron are shown in Fig. 5 and 6. Electrooxidation of hypophosphite in the solution without Ni^{2+} ions began at approximately −0.83 and −0.81V on electropolished and cathodically polarized iron, respectively. This was indicated by a deflection of the cathodic curves at these potentials, followed by the appearance of anodic current loops. In the solution containing Ni^{2+} ions, the electrooxidation of hypophosphite began at approximately −0.70V.

Permeation of hydrogen through the specimen's surface caused an increase in anodic current in the passive region, but did not affect the electrooxidation of hypophosphite. Cathodic polarization of specimens prior to testing also increased the anodic current in the passive region, without affecting the electrooxidation of hypophosphite (Fig. 6).

Voltammograms on nickel and nickel-phosphorus in various solutions are shown in Fig. 7-10. The curves were measured after cathodic polarization labeled "c.p." of specimens in 0.1M H_2SO_4. These voltammograms show that the nickel-phosphorus deposits were significantly more active than pure nickel. The higher activity of the deposits was indicated by higher anodic currents at noble potentials in all solutions, by the occurrence of a peak at around −0.78V in the plating solution, and by a steeper rise in cathodic currents at approximately −0.75V in the supporting solution with Ni^{2+} ions.

As in the case for iron, cathodic polarization of nickel and nickel-phosphorus in 0.1M H_2SO_4 before testing considerably increased anodic currents at noble potentials. At more active potentials, current increases were not significant. The effect of cathodic polarization was evident in essentially the same potential range for both materials.

Discontinuities in the voltammograms measured in the solutions without NaH_2PO_2 (Fig. 8 and 9) indicate mixed rest potentials. Currents measured noble to these potentials represent anodic dissolution of the substrates. The voltammograms show that the anodic dissolution rate of nickel-phosphorus was higher than that of nickel and that cathodic prepolarization increased the rate of dissolution of both substrates.

At potentials noble to −0.5V, anodic currents measured in the hypophosphite containing solutions (Fig. 7 and 10) were virtually identical to those measured in the hypophosphite-free solutions (Fig. 8 and 9)

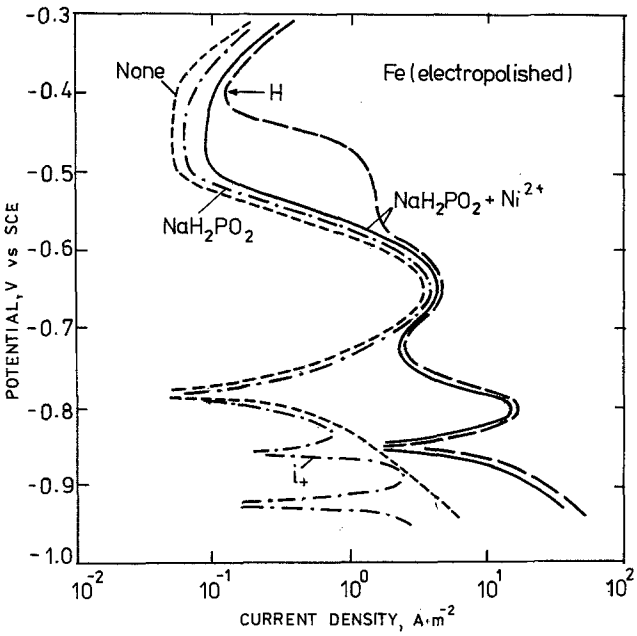


Fig. 5. Voltammograms for electropolished iron in the supporting solution with and without other components (cf. Table I), beginning at −0.30V. An arrow with "H" indicates the start of hydrogen permeation through an iron membrane. Below the discontinuities the current is cathodic, except where anodic current is denoted by i_+.

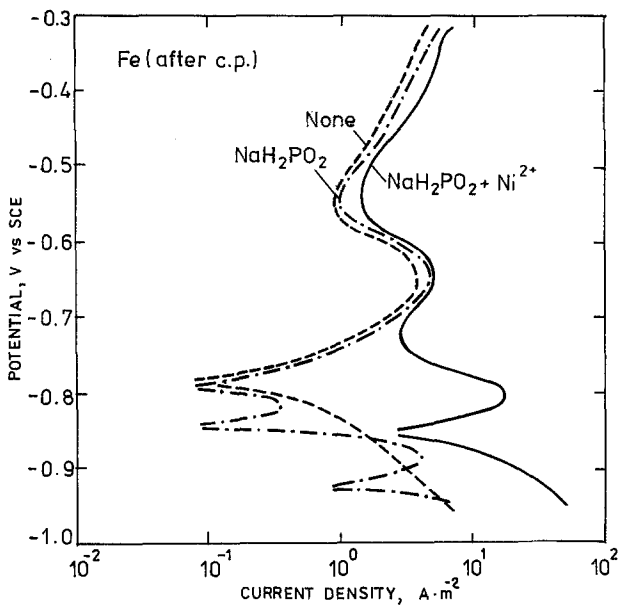


Fig. 6. Voltammograms for iron after cathodic polarization in 0.1M H_2SO_4, measured in the supporting solution with and without other components, beginning at −0.3V.

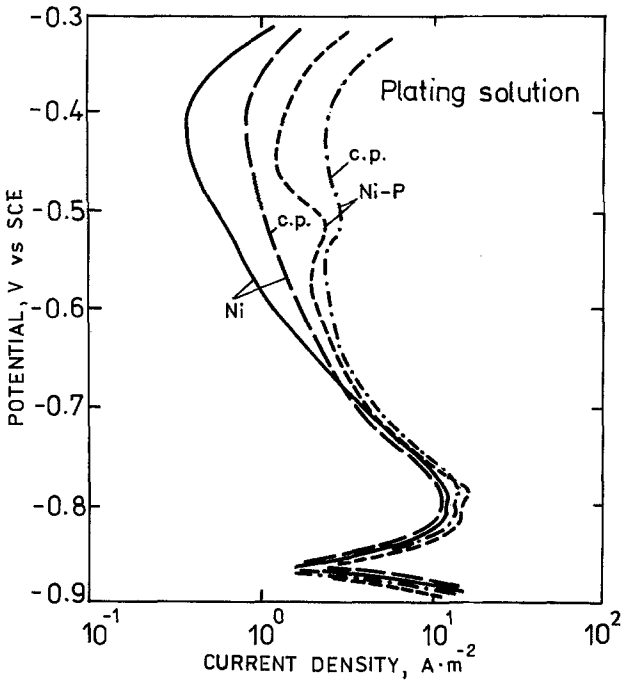


Fig. 7. Voltammograms for nickel and nickel-phosphorus deposits in the plating solution, beginning at −0.3V. Curves denoted by "c.p." were measured after cathodic polarization in 0.1M H_2SO_4.

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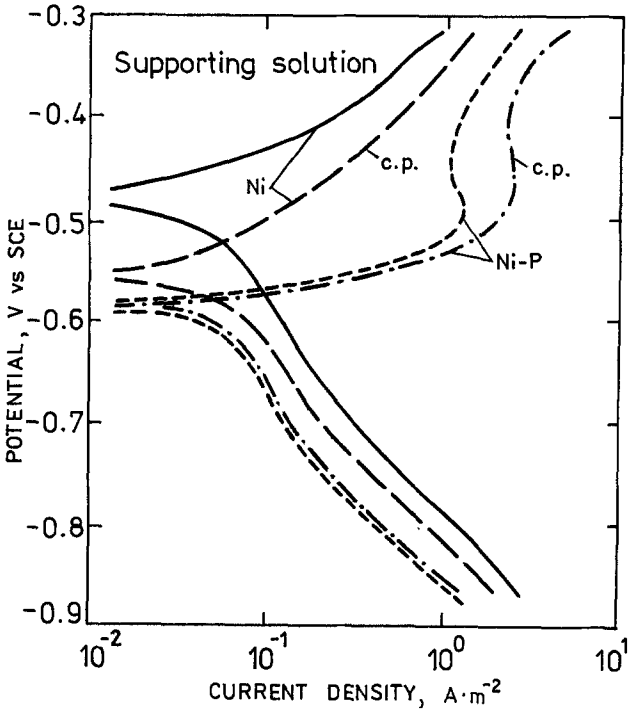


Fig. 8. Voltammograms for nickel and nickel-phosphorus deposits in the supporting solution.

on each substrate. This suggests that at these potentials the measured current was associated primarily with anodic dissolution of the substrates, and that the electrooxidation of hypophosphite either did not take place or was negligible. An anodic current measured at more active potentials was evidently associated with electrooxidation of hypophosphite. This shows that the hypophosphite electrooxidation became significant at the potentials corresponding to rest potentials in the hypophosphite-free solutions. The higher anodic dissolution current corresponds to higher electrooxidation current near the corrosion potentials.

In the plating solution (Fig. 7) the electrooxidation current of hypophosphite, active to −0.75V, reached

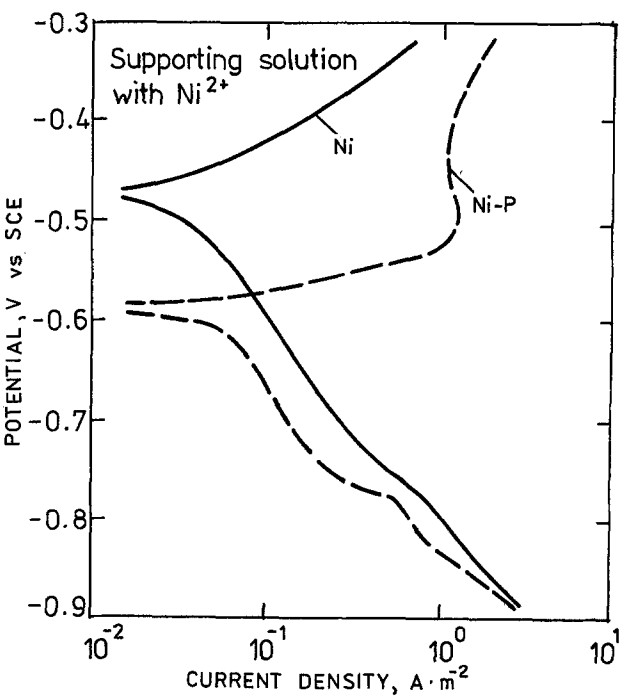


Fig. 9. Voltammograms for nickel and nickel-phosphorus deposits in the supporting solution with added Ni^{2+} ions.

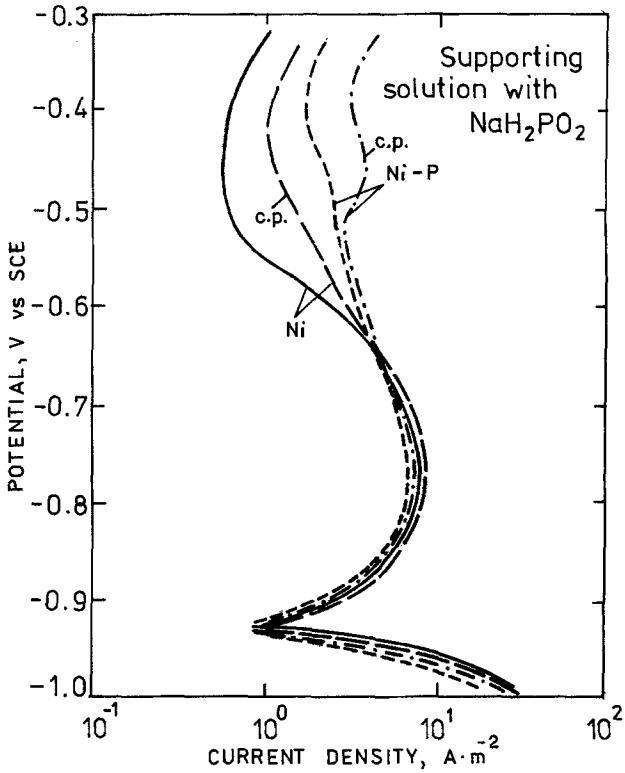


Fig. 10. Voltammograms for nickel and nickel-phosphorus deposits in the supporting solution with added NaH_2PO_2.

higher values than in the solution without Ni^{2+} ions (Fig. 10). On nickel-phosphorus deposits, in the plating solution, the increase in the current active to −0.75V was greater than on pure nickel and showed a distinct peak at −0.78V. Nickel showed only a slight tendency to form a peak at this potential.

The increase in the electrooxidation current in the plating solution active to −0.75V coincided with the start of electrolytic deposition of nickel on the substrates. The latter can be seen from cathodic curves in the supporting solution with added Ni^{2+} ions (Fig. 9). The increased electrooxidation current in the plating solution thus results from electrolytic deposition of nickel on the substrates.

Discussion

Measurements of open-circuit potentials show that the incubation time for electroless nickel plating on the respective substrates decreased in the sequence: iron > nickel > nickel-phosphorus. Oxidation of specimens prior to immersion increased the incubation time, while cathodic polarization decreased it. On iron, the incubation time was also lowered by saturation of the surface with hydrogen as indicated by permeation plateaus.

The major part of the incubation times was associated with potential arrests in the range of −0.45V for nickel and of −0.5V for iron. Voltammograms in the supporting electrolyte showed that the potential arrest for iron was in the passive-active region, while that for nickel was approximately at the corrosion potential (Fig. 8). This suggests that these potential arrests are associated with oxide dissolution.

The decrease in incubation time resulting from cathodic prepolarization or by permeation of hydrogen can also be attributed mainly to the reduction of surface oxides. This is evidenced by higher anodic currents in the passive region on iron (Fig. 5 and 6), and also on nickel and nickel-phosphorus above their corrosion potentials (Fig. 8).

Cathodic polarization leads to the reduction of surface oxides and presumably saturates the specimen surfaces with hydrogen. This absorbed hydrogen may

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contribute to reduction of oxides or hydroxides which form thereafter, especially upon immersion into the plating solution. Formation of oxides or hydroxides on the substrates in the plating solution can be surmised from the values of pH and of initial potentials, which indicate thermodynamic stability for oxides or hydroxides (16).

The voltammograms show that electrooxidation of hypophosphite started at potentials at which oxides should be reduced. On iron, this occurred at *ca.* −0.82V, which is approximately the equilibrium potential of the $Fe(OH)_2/Fe$ reaction (16); while on nickel and nickel-phosphorus this occurred at approximately the corrosion potentials. At this potential, surface oxides should be reduced to a great extent, though not completely (17-19). The effects of hydrogen and of cathodic prepolarization on the initiation of plating and on the hypophosphite electrooxidation are associated with the reduction of surface oxides. The hypophosphite electrooxidation was not affected at more active potentials. Nor was the hypophosphite electrooxidation affected by evolution of hydrogen at *ca.* −0.7V, as has been reported for palladium (13).

The greater efficiency of the nickel-phosphorus substrate (compared to that of nickel) can also be explained in terms of surface oxides. Larger anodic dissolution currents on nickel-phosphorus than those on nickel indicate that the oxide layers on nickel-phosphorus were less protective, which may result from thinner, less stable, or discontinuous oxides. Thus, the incubation time for nickel-phosphorus was shorter and hypophosphite electrooxidation in the noble potential region was greater than are observed for nickel. At potentials active to −0.65V in the solution without Ni^{2+} ions, hypophosphite electrooxidation was essentially identical for nickel and for nickel-phosphorus. This indicates that the intrinsic catalytic activity of the two substrates is nearly the same. The difference in the activity occurs essentially because of the presence of surface oxides and can be attributed to the difference in the protective properties of the oxide layers on these substrates. The explanation of Epifanova *et al.* (10), which invokes a catalytic action of phosphorus in electroless deposits, does not seem to be appropriate for the present case. The surface of nickel-phosphorus deposits is probably less contaminated with residual oxides or other species at more negative potentials. This explains the steeper rise in the electrodeposition current of nickel at −0.75V on these deposits than on nickel (Fig. 9). It is suggested that faster electrodeposition of fresh nickel on nickel-phosphorus deposits from the plating solution is responsible for the higher electrooxidation current on these deposits below −0.75V (Fig. 7). The freshly electrodeposited nickel should possess high catalytic activity owing to low contamination of its surface.

Electrolytic deposition of nickel from a plating solution (active to −0.75V) enhanced the hypophosphite electrooxidation on nickel and nickel-phosphorus, whereas on iron it initiated the hypophosphite electrooxidation at approximately this potential (Fig. 5 and 6). In the solution without Ni^{2+} ions, the hypophosphite electrooxidation on iron began at −0.81V or below. This shows that for iron in the plating solution, electrolytic deposition of nickel occurs during the spontaneous potential shift, followed by electrooxidation of hypophosphite. This is in accord with the conclusion of Gutzeit (2).

Conclusions

Cathodic polarization of iron, nickel, and nickel-phosphorus in 0.1M H_2SO_4 before testing considerably diminished incubation times for electroless nickel plating on these substrates. A similar effect was caused by permeation of hydrogen through the iron surface.

Potential scans from −0.3 to −0.92V *vs.* SCE showed that cathodic polarization and/or the presence of hydrogen in the substrates increased anodic dissolution of the substrates in the passive region, and increased the hypophosphite electrooxidation on nickel and nickel-phosphorus in the potential range of thermodynamic stability of nickel oxides or hydroxides.

The activating effect of cathodic polarization and hydrogen can be entirely ascribed to the reduction of surface oxides.

Nickel-phosphorus deposits showed lower incubation times for electroless plating of nickel, greater anodic dissolution in the passive region, and greater hypophosphite electrooxidation in the presence of oxides than those observed for nickel. This is attributed to lower protectiveness of surface oxides on these deposits.

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